

SUPPLEMENT TO “WHO CHOOSES AND WHO BENEFITS? THE DESIGN OF PUBLIC
SCHOOL CHOICE SYSTEMS”

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APPENDIX A: ADDITIONAL DATA DETAILS

A.1. *Additional Choice Options in LAUSD*

This appendix provides institutional detail on key LAUSD choice program types referenced in the paper and clarifies how each appears in our administrative records. Our main analysis uses LAUSD data from 2004–2017 and focuses on middle-school entry; within this window, we observe applications to Dual Language programs and therefore include Dual Language options in the descriptive participation analysis. However, the lottery offer records needed for the reduced-form lottery analysis and the structural estimates in Sections 4–8 are not available for Dual Language programs (and affiliated charter schools) until 2019, outside our analysis window; accordingly, these program types are excluded from Sections 4–8. Schools for Advanced Studies (SAS) are part of LAUSD’s broader landscape but operate within resident schools and typically place eligible in-zone students automatically rather than through an opt-in application and lottery process, so they are not directly comparable to the opt-in choice programs we study. We now turn to a detailed examination of the prominent alternative choice schooling options and how they relate to the education landscape in Los Angeles. Although this discussion is specific to LAUSD, the varied offerings mirror those of numerous other school districts that have embraced the portfolio management model (Hill, 2013, Hill and Campbell, 2011), with distinctions, among others, arising in how school districts organize assignment of students to schools.

Magnet Schools: LAUSD’s first foray into providing alternatives to neighborhood schools came with its magnet school program, launched in 1982 as part of a court-ordered desegregation plan. In the landmark *Crawford v. Board of Education* case, California courts found LAUSD had an obligation to alleviate school segregation. After mandatory busing was largely halted in 1981 due to a statewide ballot measure limiting state courts’ power to mandate busing, the district turned to voluntary integration programs.

LAUSD’s magnet program crept along modestly through the 1980s and 1990s—going from 42 schools at inception to roughly 150 by 2000—but the 2010s saw a rush to expand magnet offerings as charter schools began to chip away at district enrollment. Confronted with dozens of lean, new charters in underserved neighborhoods, the district pivoted to more than double its magnet offerings, from roughly 160 programs in 2008 to over 320 by 2018. Packaged into a new unified enrollment platform, this was a deliberate strategic countermove to reclaim families drawn to the rapidly expanding charter sector (Los Angeles Unified School District, 2022, Kohli et al., 2016). LAUSD magnets are explicitly organized around “theme-based” instructional programs (e.g., Science/Technology/Engineering/Math or Visual, Centers for Enriched Studies, or Visual and Performing Arts).

Affiliated Charter Schools: LAUSD distinguishes between independent charters and affiliated charters, a model unique to the district’s policy structure. Independent charters are fully autonomous non-profits with their own boards and that run their own admissions processes in accordance with state-mandated lottery provisions. Affiliated charters, in contrast, are semi-autonomous LAUSD schools. They are usually regular neighborhood schools that converted to charter status but remain under district governance, with teaching staff still maintaining ties to the local teacher union. Affiliated charters generally continue to serve their neighborhood attendance area first and then use a lottery for any open seats for out-of-area students, effectively behaving like district schools with expanded autonomy.

LAUSD’s affiliated charter sector evolved from a handful of early conversions in the 1990s and early 2000s into a core pillar of the district’s choice portfolio by the late 2010s. What began

with isolated cases—such as Palisades Charter High School’s conversion in 1993—accelerated around 2009, when district leaders embraced charter conversion as a tactic to stem student losses to competing independent charter schools. Over the ensuing decade, LAUSD authorized the conversion of dozens of neighborhood campuses into affiliated charters. The district now offers 52 affiliated charters, separate from the roughly 150 charter schools they serve as the authorizing body for.

Dual Language Programs: Dual Language Education (DLE) schools offer grade-level instruction, biliteracy development, and sociocultural learning to students in two languages: English and a target language. LAUSD launched its first Spanish-English dual immersion program in 1991, followed by a Korean-English program in 1992. The programs gained steady momentum through the 1990s and 2000s, weathering significant challenges including California’s English-only movement under Proposition 227 in 1998, which severely restricted bilingual education statewide. Despite these constraints, LAUSD has maintained a commitment to dual language instruction, growing from 13 Two-Way Dual Language programs in 2005 to 146 by 2022. This persistence positioned the district to rapidly expand when California voters passed Proposition 58—a repeal of Proposition 227—in 2016, restoring flexibility for multilingual programming. Currently, LAUSD offers more than 230 DLE programs.

The district operates three distinct program models: Two-Way Immersion (TWI) programs that integrate native English speakers with native speakers of the target language; One-Way Immersion (OWI) programs designed specifically for English Learners whose home language matches the target language; and World Language Immersion (WLI) programs that provide English-speaking students access to second language instruction. DLE programming encompasses seven languages, Arabic, Armenian, French, Japanese, Korean, Mandarin, and Spanish, with Spanish programs comprising the largest share due to both demographic demand and the district’s substantial Latino enrollment. An overwhelming majority of these programs are elementary school programs, even though in recent years there has been some minor expansion into the secondary grade levels.

Schools for Advanced Studies: LAUSD’s Schools for Advanced Studies (SAS) initiative began as a district-level strategy to recognize and expand exemplary gifted and talented education (GATE) at resident (i.e. non-magnet, non-charter) schools. The rationale was that many high-ability or gifted students were being underserved by the standard GATE framework, and that by designating certain schools as SAS “demonstration sites,” LAUSD could concentrate resources around models that might then influence practices across the district. Importantly, magnet schools, independent charters, and dual-language programs are explicitly ineligible for SAS designation, so SAS remains rooted in the resident school infrastructure.

Within LAUSD’s broader choice ecosystem, SAS programs satisfy a distinctive niche. Unlike magnet programs, SAS was not born of desegregation court orders or integration mandates; its purpose is not explicitly to draw students from across zones (though cross-boundary permitting is possible in some cases) but rather to strengthen the quality of gifted education within neighborhood schools. In practice this means SAS cannot be lumped with magnets or charters without obscuring key differences: SAS sites remain “resident schools first,” meaning that students living in their boundaries who qualify for GATE/SAS are typically placed automatically (unless opting out).

APPENDIX B: OVERSUBSCRIBED SCHOOL LOTTERY DETAILS

B.1. *Choice School Lottery Offers*

When applications to non-neighborhood schools (e.g., magnets and other specialized programs) exceed available seats, LAUSD allocates seats by lottery. Lotteries are run within strata defined by the application year, student grade, school/program, race, and priority-points level, so applicants only compete against others in the same stratum. The data available for our analysis report each applicant's randomly assigned number, whether the applicant ultimately received an offer from the program, and all fields needed to define the lottery strata. In this system, higher lottery numbers correspond to higher priority. Programs set an undisclosed cutoff lottery number such that applicants with lottery numbers at or above the cutoff are eligible for initial offers, while those below the cutoff are not; applicants on the offer side of this cutoff receive initial offers. As seats open, programs may also extend offers to waitlisted students and, in a small number of cases, make post-lottery offers to late applicants if capacity remains.

For the analysis, we construct an indicator for having a random number on the offer side of the cutoff. Because cutoffs are not recorded, we infer them from the joint distribution of random numbers and final offers, following [Porter and Yu \(2015\)](#), who extend RD methods to cases with unknown discontinuity points. Concretely, we estimate the threshold that maximizes the discontinuity in the offer probability.

Identification of Potential Lotteries: Lotteries occur at the intersection of year, school, grade, race, and priority points levels. We first identify which of these cells experienced lottery-based assignment using the following procedure. We begin by calculating offers and applications. For each combination of application year, student grade, school, race, and priority points level, we compute the total number of offers made and the total number of applications received. Next, we identify oversubscribed cells by flagging cells as oversubscribed when applications exceed offers. We then define potential lottery cells. A cell represents a potential lottery if it satisfies three conditions: it has multiple applicants, is oversubscribed, and has at least one offer made. Each unique combination of year, school, grade, race, and priority points level that meets the potential lottery criteria is considered a lottery.

Cutoff Estimation: For each potential lottery identified above, we estimate the lottery number cutoff using a regression-based approach that searches for the point of maximum discontinuity in assignment probability. Within each lottery cell, we rank applicants by their lottery number. We exclude the minimum and maximum values to avoid boundary issues. For each potential cutoff rank c , we create an indicator variable:

$$\text{above}_c = 1[\text{lottery_number} \geq c]. \quad (1)$$

This indicator equals one if an applicant's lottery number is at or above the candidate cutoff (i.e., on the high-number, offer-eligible side of the cutoff). Then, we estimate the discontinuity at each candidate cutoff. Formally, the specification is:

$$\text{offer_magnet}_i = \alpha + \beta \cdot \text{above}_c + \epsilon_i. \quad (2)$$

The coefficient β captures the change in probability of receiving an offer at the candidate cutoff c . Standard errors are heteroskedasticity robust. We select the cutoff value c that maximizes the R^2 among all candidates satisfying:

- $\beta > 0$ (offers increase above the cutoff)
- $p\text{-value} \leq 0.01$ (statistically significant at the 1% level) or $p\text{-value}$ is missing (which occurs when $\beta = 1$ with perfect prediction)
- Model is identified (rank condition satisfied)

APPENDIX C: DEMAND MODEL

This appendix section provides technical details related to the estimation of the model, primarily discussing the simulated likelihood function and its associated gradient.

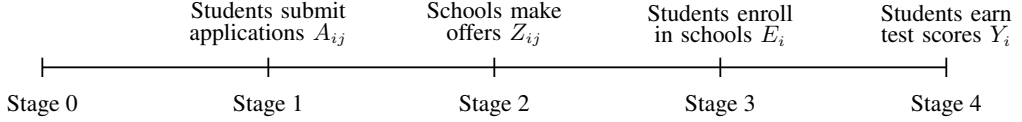


FIGURE C.1.—Stages of Decisions and Outcomes in the Model. *Note:* This figure summarizes key decisions and endogenous variables in our model.

C.1. Overview

Section 7 of the main paper outlines the model and gives its likelihood function in Equation (9), which depends on the parameter vector

$$\Omega = (\{\delta_j\}, \beta_x, \psi_D, \psi_{D2}, \mu_c, \mu_{cx}, \{\mu_k, \sigma_k, p_k\}, \sigma_\eta).$$

Note that $\omega \in \Omega$ denotes an arbitrary element of the parameter vector. This section provides details associated with the likelihood maximization procedure. Likelihoods are calculated via simulation, where we use $r = 1, \dots, R$ as the index for simulation draws.

In practice, we maximize the following function:

$$\mathcal{L} = \sum_{i=1}^N \ln \bar{P}_i(\Omega), \quad (3)$$

where the term $\bar{P}_i(\Omega) = (1/R) \sum_{r=1}^R P_{ir}(\Omega)$ is the simulated likelihood contribution for individual i .

The partial derivative of the likelihood function with respect to parameter ω is given by:

$$\frac{\partial \mathcal{L}}{\partial \omega} = \sum_{i=1}^N \frac{1}{\bar{P}_i(\Omega)} \underbrace{\left(\frac{1}{R} \sum_{r=1}^R \frac{\partial \ln P_{ir}(\Omega)}{\partial \omega} \cdot P_{ir}(\Omega) \right)}_{\bar{g}_i(\Omega)}, \quad (4)$$

where $\bar{g}_i(\omega)$ is the average partial derivative for individual i , and the average is calculated across simulation draws. Therefore, for a given parameter ω , the partial derivative of the likelihood can be written as:

$$\frac{\partial \mathcal{L}}{\partial \omega} = \sum_{i=1}^N \frac{\bar{g}_i(\Omega)}{\bar{P}_i(\Omega)}.$$

C.1.1. Log-Likelihood and Gradient

For individual i and simulation draw r , the per-draw log-likelihood is defined as:¹

$$\ell_{ir}(\Omega) \equiv \ln P_{ir}(\Omega) = \ln P_{A,ir}(A_i | \Omega) + \ln P_{E,ir}(E_i | \Omega), \quad (5)$$

where A_i is an application vector and E_i is the school that a child enrolls in. For parsimony, we suppress the conditioning on Ω in the expressions that follow.

Recall that Equation (7) in the main paper defines the following term:

$$P_{A,ir}(A_i) \approx \frac{\exp(V_{ir}(A_i)/\lambda)}{\sum_{a'} \exp(V_{ir}(a')/\lambda)},$$

where the term $V_{ir}(A_i) \equiv e_{ir}(A_i) - c_{ir}(A_i)$ is expected application utility net of the corresponding application cost, as defined in the main paper. In addition, Equation (8) in the main paper defines the enrollment probability:

$$P_{E,ir}(E_i = j) = \frac{\exp(v_{ijr})}{\sum_{j' \in \mathcal{O}_i} \exp(v_{ij'r})},$$

where E_i is the school in which a student enrolls, $\mathcal{O}_i$ is the school offer set, and the systematic utility term is defined as $v_{ijr} = \delta_j + \theta_{ir} + \beta_c X_i - \psi_D D_{ij} - \psi_{D2} D_{ij}^2 - \psi_{Dx}(D_{ij} \times X_i)$.

Here, $P_{A,ir}$ is the portion of the likelihood contribution that comes from the application stage, and $P_{E,ir}$ is the portion of the likelihood contribution that comes from the enrollment stage. At the application stage, families are weighing their expected utilities from an application, $e_{ir}(A_i)$, against the costs from applying, $c_{ir}(A_i)$, which we denote by V_{ir} in the notation above. The kernel smoothing factor λ allows us to convert the application choice into a multinomial logit problem following (Walters, 2018).

Returning to Equation 5, we know that the per-draw log-likelihood for individual i is:

$$\ell_{ir}(\Omega) = \ln P_{ir}(\Omega).$$

The partial derivative with respect to parameter ω , conditional on a draw r , is:

$$\frac{\partial \ell_{ir}}{\partial \omega} = \frac{1}{\lambda} \left[\frac{\partial V_{ir}(A_i)}{\partial \omega} - \sum_a P_{A,ir}(a) \frac{\partial V_{ir}(a)}{\partial \omega} \right] + \frac{\partial \ln P_{E,ir}(E_i)}{\partial \omega}. \quad (6)$$

Note that A_i denotes the application vector submitted by i in data, and a denotes one of the possible application vectors that could be submitted by i .

¹Conditioning on data such as X_i, D_i, θ_i is suppressed from the following notation.

The per-draw gradient vector $g_{ir}(\Omega)$ stacks these partial derivatives and has M -many entries based on the dimension of Ω :

$$g_{ir}(\Omega) = \begin{pmatrix} \frac{\partial \ell_{ir}}{\partial \omega_1} \\ \vdots \\ \frac{\partial \ell_{ir}}{\partial \omega_m} \\ \vdots \\ \frac{\partial \ell_{ir}}{\partial \omega_M} \end{pmatrix},$$

where the subscript m is the parameter index.

For each element of Ω , Equation (6) implies a decomposition into application and enrollment components. We therefore define $g_{ir}^A(\Omega)$ and $g_{ir}^E(\Omega)$ so that we have:

$$g_{ir}(\Omega) = g_{ir}^A(\Omega) + g_{ir}^E(\Omega),$$

where the m -th entries of $g_{ir}^A(\Omega)$ and $g_{ir}^E(\Omega)$ correspond to the first and second terms on the right-hand side of Equation (6), respectively. In this case, the simulated gradient contribution of individual i is:

$$\bar{g}_i(\Omega) = \frac{1}{R} \sum_{r=1}^R (g_{ir}^A(\Omega) + g_{ir}^E(\Omega)).$$

C.1.2. Application-Stage Gradient

For any feasible application vector a , we have that $V_{ir}(a) = e_{ir}(a) - c_{ir}(a)$. The partial derivative with respect to ω_m is:

$$\frac{\partial V_{ir}(a)}{\partial \omega_m} = \frac{\partial e_{ir}(a)}{\partial \omega_m} - \frac{\partial c_{ir}(a)}{\partial \omega_m}.$$

Based on Equation (6), we denote the m -th component of the application-stage gradient vector $g_{ir}^A(\Omega)$ as $g_{ir,m}^A$. This can be written as:

$$g_{ir,m}^A = \frac{1}{\lambda} \left[\frac{\partial e_{ir}(A_i)}{\partial \omega_m} - \sum_a P_{A,ir}(a) \frac{\partial e_{ir}(a)}{\partial \omega_m} \right] - \frac{1}{\lambda} \left[\frac{\partial c_{ir}(A_i)}{\partial \omega_m} - \sum_a P_{A,ir}(a) \frac{\partial c_{ir}(a)}{\partial \omega_m} \right], \quad (7)$$

where A_i denotes the application vector submitted by i in the data.

Note that if an individual chooses *not* to apply, then $V_{ir}(A_i) = 0$, so the terms involving $\partial e_{ir}(A_i)/\partial \omega_m$ and $\partial c_{ir}(A_i)/\partial \omega_m$ drop out. In that case,

$$g_{ir,m}^A = -\frac{1}{\lambda} \left[\sum_a P_{A,ir}(a) \frac{\partial e_{ir}(a)}{\partial \omega_m} \right] + \frac{1}{\lambda} \left[\sum_a P_{A,ir}(a) \frac{\partial c_{ir}(a)}{\partial \omega_m} \right]. \quad (8)$$

C.1.3. Enrollment-Stage Gradient

Based on Equation (8) in the main paper, we can write the probability of enrolling at school j as:

$$P_{E,ir}(E_i = j) = \frac{\exp(v_{ijr})}{\sum_{j' \in \mathcal{O}_i} \exp(v_{ij'r})},$$

where j' is an offered school, and $\mathcal{O}_i$ denotes the offer set for student i . Our specification for utility assumes

$$v_{ijr} = \delta_j + \theta_{ir} + \beta_c X_i - \psi_D D_{ij} - \psi_{D2} D_{ij}^2 - \psi_{Dx} (D_{ij} \times X_i).$$

Note that the outside option is included in $\mathcal{O}_i$ and normalized appropriately.²

For individual i and draw r , the enrollment-stage per-draw log-likelihood can be written as

$$\begin{aligned} \ell_{ir}^E(\Omega) &= \ln P_{E,ir}(E_i | \Omega) \\ &= \ln \left(\frac{\exp(v_{iEr})}{\sum_{j' \in \mathcal{O}_i} \exp(v_{ij'r})} \right) \\ &= v_{iEr} - \ln \left(\sum_{j' \in \mathcal{O}_i} \exp(v_{ij'r}) \right). \end{aligned}$$

The first term depends only on the utility index for the school that student i actually enrolls in (the observed $E_i = j$), while the second term depends on the utilities of all schools in the offer set $\mathcal{O}_i$.

For a scalar parameter $\omega_m \in \Omega$, the derivative of ℓ_{ir}^E is therefore:

$$\begin{aligned} \frac{\partial \ell_{ir}^E}{\partial \omega_m} &= \frac{\partial v_{iEr}}{\partial \omega_m} - \frac{\partial}{\partial \omega_m} \ln \left(\sum_{j' \in \mathcal{O}_i} \exp(v_{ij'r}) \right) \\ &= \frac{\partial v_{iEr}}{\partial \omega_m} - \frac{1}{\sum_{j' \in \mathcal{O}_i} \exp(v_{ij'r})} \sum_{j' \in \mathcal{O}_i} \exp(v_{ij'r}) \frac{\partial v_{ij'r}}{\partial \omega_m}. \end{aligned}$$

We can simplify this expression by noting that the first term can be written as a sum over the offer set:

$$\frac{\partial v_{iEr}}{\partial \omega_m} = \sum_{j' \in \mathcal{O}_i} \mathbf{1}\{E_i = j'\} \frac{\partial v_{ij'r}}{\partial \omega_m},$$

²For students without any offer, we treat enrollment as degenerate on the outside option; in that case the enrollment-stage log-likelihood contribution is constant and the gradient is zero.

where the indicator term is only non-zero when the index of the summation takes the value of the school that a child enrolls in (i.e., $j' = j$). It will also be helpful to define the following expression:

$$P_{E,ir}(j) \equiv \frac{\exp(v_{ijr})}{\sum_{j' \in \mathcal{O}_i} \exp(v_{ij'r})}.$$

Returning to the partial derivative, we can use the immediately preceding results to simplify:

$$\begin{aligned} \frac{\partial \ell_{ir}^E}{\partial \omega_m} &= \sum_{j' \in \mathcal{O}_i} \mathbf{1}\{E_i = j'\} \frac{\partial v_{ij'r}}{\partial \omega_m} - \sum_{j' \in \mathcal{O}_i} P_{E,ir}(j') \frac{\partial v_{ij'r}}{\partial \omega_m} \\ &= \sum_{j' \in \mathcal{O}_i} \left(\mathbf{1}\{E_i = j'\} - P_{E,ir}(j') \right) \frac{\partial v_{ij'r}}{\partial \omega_m}. \end{aligned} \quad (9)$$

The gradient vector $g_{ir}^E(\Omega)$ stacks these partial derivatives, with the m -th component being:

$$g_{ir,m}^E(\Omega) \equiv \frac{\partial \ell_{ir}^E}{\partial \omega_m}.$$

To more fully consider these expressions, consider school s and calculate $\partial \ell_{ir}^E / \partial \delta_s$. Since the parameter δ_s enters only the utility index for school s , we know that:

$$\frac{\partial v_{ij'r}}{\partial \delta_s} = \mathbf{1}\{j' = s\} \quad \text{for each } j' \in \mathcal{O}_i.$$

Substituting into Equation (9) yields:

$$\begin{aligned} g_{ir,\delta_s}^E &= \frac{\partial \ell_{ir}^E}{\partial \delta_s} = \sum_{j' \in \mathcal{O}_i} \left(\mathbf{1}\{E_i = j'\} - P_{E,ir}(j') \right) \mathbf{1}\{j' = s\} \\ &= \mathbf{1}\{E_i = s\} - P_{E,ir}(s). \end{aligned}$$

where only the term with $j' = s$ survives in the sum.

More generally, when a parameter ω_m enters v_{ijr} linearly and only for school s , we have

$$g_{ir,\omega_m}^E = \frac{\partial \ell_{ir}^E}{\partial \omega_m} = \left(\mathbf{1}\{E_i = s\} - P_{E,ir}(s) \right) \frac{\partial v_{isr}}{\partial \omega_m}, \quad (10)$$

where the derivative $\partial v_{isr} / \partial \omega_m$ is given by the corresponding covariate or transformation. For example, consider the cases:

$$\frac{\partial v_{isr}}{\partial \psi_D} = -D_{is}, \quad \frac{\partial v_{isr}}{\partial \psi_{D2}} = -D_{is}^2,$$

and the corresponding gradient components follow from Equation (10).

Finally, if a student receives no offer and enrolls in the outside option by construction, then $P_{E,ir}(E_i = 0)$ is identically one and does not depend on Ω . In this case, the enrollment-stage gradient contribution is:

$$\frac{\partial \ell_{ir}^E}{\partial \omega_m} = 0 \quad \text{for all } \omega_m.$$

C.1.4. *Estimating a Mixture of Normals for Idiosyncratic Choice School Preferences*

The likelihood contribution of individual i is:

$$\mathcal{L}(A_i, Z_i, E_i \mid X_i, D_i) = \int P_A(A_i \mid X_i, D_i, \theta, \eta) \times P_E(E_i \mid Z_i, X_i, D_i, \theta) dF(\theta, \eta \mid X_i, D_i),$$

where it is implicit that we must integrate out the unobserved taste and cost shocks θ_i and η_i (we suppress the i subscript on (θ, η) in the notation below).

In a model with a single normal component, we would assume $\theta \sim N(0, \sigma_\theta^2)$ and $\eta \sim N(0, \sigma_\eta^2)$, independent of (X_i, D_i) , so

$$dF(\theta, \eta \mid X_i, D_i) = \phi(\theta; 0, \sigma_\theta^2) \phi(\eta; 0, \sigma_\eta^2) d\theta d\eta,$$

where $\phi(x; \mu, \sigma^2)$ denotes a normal density for x with mean μ and variance σ^2 .

To allow for more flexible heterogeneity in tastes for choice schools, we generalize this to a finite mixture of normals. In the mixture model with K types, we have:

$$\theta \sim \sum_{k=1}^K p_k N(\mu_k, \sigma_k^2),$$

where p_k is the share of type k in the population and each type k has its own mean μ_k and standard deviation σ_k . We maintain a mean-zero normalization for θ ,

$$E[\theta] = \sum_{k=1}^K p_k \mu_k = 0,$$

to pin down the overall location of the taste distribution and avoid confounding the mean of θ_i with the school fixed effects $\{\delta_j\}$ in the utility index. This restriction imposes one linear constraint across the K component means. For example, we can express:

$$\mu_K = -\frac{\sum_{j=1}^{K-1} p_j \mu_j}{p_K},$$

so that there are $K - 1$ unrestricted mean parameters. We also estimate K standard deviations σ_k for each type. Keeping the assumptions on the distribution of η constant, the joint density becomes:

$$dF(\theta, \eta \mid X_i, D_i) = \left(\sum_{k=1}^K p_k \phi(\theta; \mu_k, \sigma_k^2) \right) \phi(\eta; 0, \sigma_\eta^2) d\theta d\eta.$$

Under this mixture specification, the likelihood contribution of individual i is:

$$\mathcal{L}(A_i, Z_i, E_i \mid X_i, D_i) = \sum_{k=1}^K p_k \left(\int P_A(A_i \mid X_i, D_i, \theta, \eta) \times P_E(E_i \mid Z_i, X_i, D_i, \theta) \phi(\theta; \mu_k, \sigma_k^2) \phi(\eta; 0, \sigma_\eta^2) d\theta d\eta \right).$$

In the $K > 1$ case, we must also parameterize the type probabilities p_k . These probabilities must satisfy $p_k \geq 0$ for all k and $\sum_{k=1}^K p_k = 1$. Rather than estimate the p_k directly under these constraints, we follow a standard approach and work with an unconstrained parameter vector $(\alpha_1, \dots, \alpha_K) \in \mathbb{R}^K$ and map it into probabilities using the soft-max (multinomial logit) transformation:

$$p_k = \frac{\exp(\alpha_k)}{\sum_{k'=1}^K \exp(\alpha_{k'})}, \quad k = 1, \dots, K$$

This guarantees that each p_k is strictly positive and that the probabilities sum to one for any values of $(\alpha_1, \dots, \alpha_K)$.

The soft-max transformation is invariant to adding a constant to all components; that is, $p_k(\alpha_1 + c, \dots, \alpha_K + c) = p_k(\alpha_1, \dots, \alpha_K)$ for any scalar c . To obtain an identified parameterization, we therefore normalize one component, which we take to be $\alpha_K = 0$. Under this normalization, the probabilities can be written as:

$$p_k = \frac{\exp(\alpha_k)}{1 + \sum_{k'=1}^{K-1} \exp(\alpha_{k'})},$$

and

$$p_K = \frac{1}{1 + \sum_{k'=1}^{K-1} \exp(\alpha_{k'})}.$$

Thus, there are $K - 1$ free parameters governing the type probabilities.

Combining the $K - 1$ unrestricted means, the K standard deviations, and the $K - 1$ type-probability parameters, the mixture distribution for θ is characterized by $3K - 2$ free parameters.

C.1.5. *Changes to the Gradient in the Mixture Case*

With the mixture-of-normals specification for θ , the simulated choice probability for individual i is:

$$P_i(\Omega) = \sum_{k=1}^K p_k \bar{P}_{i,k}(\Omega).$$

In this expression, the term:

$$\bar{P}_{i,k}(\Omega) = \frac{1}{R} \sum_{r=1}^R P_{ir,k}(\Omega)$$

is the simulated probability for individual i under mixture component k , averaged over R simulation draws. The sample log-likelihood is:

$$\mathcal{L}(\Omega) = \sum_{i=1}^N \ln P_i(\Omega).$$

Consider a scalar parameter $\omega \in \Omega$ that does not parameterize the type probabilities p_k (i.e., ω is not one of the $\{\alpha_k\}$, so $\partial p_k / \partial \omega = 0$). The derivative of the log-likelihood with respect to ω is:

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial \omega} &= \sum_{i=1}^N \frac{1}{P_i(\Omega)} \frac{\partial P_i(\Omega)}{\partial \omega} \\ &= \sum_{i=1}^N \frac{1}{P_i(\Omega)} \frac{\partial}{\partial \omega} \left(\sum_{k=1}^K p_k \bar{P}_{ik}(\Omega) \right) \\ &= \sum_{i=1}^N \frac{1}{P_i(\Omega)} \left(\sum_{k=1}^K p_k \frac{\partial \bar{P}_{ik}(\Omega)}{\partial \omega} \right) \\ &= \sum_{i=1}^N \frac{1}{P_i(\Omega)} \left(\sum_{k=1}^K p_k \frac{1}{R} \sum_{r=1}^R \frac{\partial P_{irk}(\Omega)}{\partial \omega} \right). \end{aligned}$$

Using the identity

$$\frac{\partial P_{irk}(\Omega)}{\partial \omega} = P_{irk}(\Omega) \frac{\partial \ln P_{irk}(\Omega)}{\partial \omega},$$

we can rewrite the inner sum as:

$$\frac{1}{R} \sum_{r=1}^R \frac{\partial P_{irk}(\Omega)}{\partial \omega} = \frac{1}{R} \sum_{r=1}^R P_{irk}(\Omega) \frac{\partial \ln P_{irk}(\Omega)}{\partial \omega} \equiv \bar{g}_{ik}(\Omega),$$

where $\bar{g}_{ik}(\Omega)$ is the simulated gradient contribution for individual i under mixture component k . Substituting back, we obtain:

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial \omega} &= \sum_{i=1}^N \frac{1}{P_i(\Omega)} \left(\sum_{k=1}^K p_k \bar{g}_{ik}(\Omega) \right) \\ &= \sum_{i=1}^N \frac{\bar{g}_i(\Omega)}{P_i(\Omega)}, \end{aligned}$$

where the term

$$\bar{g}_i(\Omega) \equiv \sum_{k=1}^K p_k \bar{g}_{ik}(\Omega)$$

is the mixture-weighted average of the K simulated gradient contributions for individual i .

Relative to the case of a single-normal model, the structure of the individual gradient contribution is qualitatively similar, but we must now keep track of the K component-specific simulated probabilities $\bar{P}_{ik}(\Omega)$ and their associated gradients $\bar{g}_{ik}(\Omega)$. The overall gradient is a probability-weighted average of these K simulated gradients for each individual.

C.1.6. Gradients with Respect to Type Probability Parameters in the Mixture Case

We now consider the gradient of the log-likelihood with respect to the type-probability parameters $\{\alpha_k\}_{k=1}^{K-1}$. Recall that the simulated probability for individual i under the mixture specification is:

$$P_i(\Omega) = \sum_{k=1}^K p_k \bar{P}_{ik}(\Omega),$$

where $p_k = p_k(\alpha_1, \dots, \alpha_{K-1})$ and $\bar{P}_{ik}(\Omega) = \frac{1}{R} \sum_{r=1}^R P_{irk}(\Omega)$ is the simulated probability for individual i under component k . The individual log-likelihood contribution is:

$$\ell_i(\Omega) \equiv \ln P_i(\Omega).$$

For a given α_k (with $k = 1, \dots, K-1$), the derivative of ℓ_i is:

$$\begin{aligned} \frac{\partial \ell_i}{\partial \alpha_k} &= \frac{1}{P_i(\Omega)} \frac{\partial P_i(\Omega)}{\partial \alpha_k} \\ &= \frac{1}{P_i(\Omega)} \frac{\partial}{\partial \alpha_k} \left(\sum_{k'=1}^K p_{k'} \bar{P}_{ik'}(\Omega) \right) \\ &= \frac{1}{P_i(\Omega)} \left(\sum_{k'=1}^K \frac{\partial p_{k'}}{\partial \alpha_k} \bar{P}_{ik'}(\Omega) + \sum_{k'=1}^K p_{k'} \frac{\partial \bar{P}_{ik'}(\Omega)}{\partial \alpha_k} \right). \end{aligned}$$

Only the mixture weights p_k and the component K contribution $\bar{P}_{iK}(\Omega)$ depend on the type-probability parameters. For $k < K$, we have $\partial \bar{P}_{ik'}(\Omega) / \partial \alpha_k = 0$, so we can simplify this expression to:

$$\frac{\partial \ell_i}{\partial \alpha_k} = \frac{1}{P_i(\Omega)} \left(\sum_{k'=1}^K \frac{\partial p_{k'}}{\partial \alpha_k} \bar{P}_{ik'}(\Omega) + p_K \frac{\partial \bar{P}_{iK}(\Omega)}{\partial \alpha_k} \right). \quad (11)$$

Derivatives of the mixture weights. To obtain $\partial p_{k'} / \partial \alpha_k$, recall that under the multinomial-logit (soft-max) parameterization with normalization $\alpha_K = 0$ we have

$$p_k = \frac{\exp(\alpha_k)}{K-1 + \sum_{k'=1}^{K-1} \exp(\alpha_{k'})} \quad \text{for } k = 1, \dots, K-1,$$

and the normalization $\alpha_K = 0$ implies

$$p_K = \frac{1}{K-1 + \sum_{k'=1}^{K-1} \exp(\alpha_{k'})}.$$

Differentiating with respect to α_k (for a fixed $k \in \{1, \dots, K-1\}$) yields:³

$$\frac{\partial p_k}{\partial \alpha_k} = p_k(1 - p_k),$$

$$\frac{\partial p_{k'}}{\partial \alpha_k} = -p_{k'}p_k \quad \text{for all } k' \neq k$$

(where the second expression applies in particular to $k' = K$).

Derivatives of the final component contribution. The next term of interest in Equation (11) captures the dependence of the K -th component contribution $\bar{P}_{iK}(\Omega)$ on α_k through the mean-zero restriction on θ . Recall that

$$\bar{P}_{iK}(\Omega) = \frac{1}{R} \sum_{r=1}^R P_{irK}(\Omega),$$

and hence:

$$\frac{\partial \bar{P}_{iK}(\Omega)}{\partial \alpha_k} = \frac{1}{R} \sum_{r=1}^R \frac{\partial P_{irK}(\Omega)}{\partial \alpha_k} = \frac{1}{R} \sum_{r=1}^R P_{irK}(\Omega) \frac{\partial \ln P_{irK}(\Omega)}{\partial \alpha_k}.$$

The derivative of the per-draw log-probability can be decomposed into application and enrollment components:

$$\frac{\partial \ln P_{irK}(\Omega)}{\partial \alpha_k} = g_{ir,K}^A(\alpha_k) + g_{ir,K}^E(\alpha_k),$$

where $g_{ir,K}^A(\alpha_k)$ and $g_{ir,K}^E(\alpha_k)$ collect the terms arising from the application and enrollment stages, respectively, when the type is K .⁴

The only channel through which α_k affects these terms is the normalized mean of the K -th type. Under the mean-zero restriction $E[\theta] = 0$ and the soft-max parameterization of the p_k ,

³These are the familiar soft-max derivatives that follow from applying the quotient rule to $p_k = \exp(\alpha_k) / (1 + \sum_{n=1}^{K-1} \exp(\alpha_n))$.

⁴As in Equation (7), $g_{ir,K}^A(\alpha_k)$ is built from derivatives of the application index $V_{ir}(a) = e_{ir}(a) - c_{ir}(a)$ with respect to α_k , and $e_{ir}(a)$ is the expected payoff from applying to portfolio a .

we can write:⁵

$$\mu_K = - \sum_{k'=1}^{K-1} \exp(\alpha_{k'}) \mu_n,$$

so that:

$$\frac{\partial \mu_K}{\partial \alpha_k} = - \exp(\alpha_k) \mu_k. \quad (12)$$

In the application-stage expected utility for type K , this implies:

$$\frac{\partial e_{ir,K}(a)}{\partial \alpha_k} = \pi_{ia} P_{E,irK} \frac{\partial \mu_K}{\partial \alpha_k} = -\pi_{ia} P_{E,irK} \exp(\alpha_k) \mu_k.$$

Since the application-cost function $c_{ir}(a)$ does not depend on the type-probability parameters $\{\alpha_k\}$, the derivative of the application-stage index $V_{ir}(a) = e_{ir}(a) - c_{ir}(a)$ with respect to α_k is entirely driven by $\partial e_{ir,K}(a)/\partial \alpha_k$.

Substituting these derivatives into the general application-stage gradient expression in Equation (7) yields $g_{ir,K}^A(\alpha_k)$. A similar substitution into the enrollment-stage gradient expression in Equation (9), using $\partial v_{ijr}/\partial \alpha_k = \partial \mu_K/\partial \alpha_k$ for type K , yields $g_{ir,K}^E(\alpha_k)$.

Putting these pieces together, we can summarize the gradient of the individual log-likelihood with respect to α_k as

$$\frac{\partial \ell_i}{\partial \alpha_k} = \frac{1}{P_i(\Omega)} \left(\sum_{k'=1}^K \frac{\partial p_{k'}}{\partial \alpha_k} \bar{P}_{ik'}(\Omega) + p_K \bar{g}_{iK}(\alpha_k) \right),$$

where the last term in the parentheses is given by:

$$\bar{g}_{iK}(\alpha_k) \equiv \frac{1}{R} \sum_{r=1}^R P_{irK}(\Omega) [g_{ir,K}^A(\alpha_k) + g_{ir,K}^E(\alpha_k)],$$

which is the simulated gradient contribution for individual i arising from the impact of α_k on the K -th mixture component.

⁵Under the soft-max parameterization with normalization $\alpha_K = 0$, we have that $\mu_K = -(\sum_{k=1}^{K-1} p_k \mu_k)/p_K$. In addition, we have:

$$p_k = \frac{\exp(\alpha_k)}{D(\alpha)} \quad (k = 1, \dots, K-1), \quad p_K = \frac{1}{D(\alpha)},$$

where $D(\alpha) \equiv 1 + \sum_{k=1}^{K-1} \exp(\alpha_k)$. Substituting these expressions into the μ_K above yields:

$$\mu_K = - \frac{\sum_{k=1}^{K-1} [\exp(\alpha_k)/D(\alpha)] \mu_k}{1/D(\alpha)} = - \sum_{k=1}^{K-1} \exp(\alpha_k) \mu_k.$$

C.1.7. Gradients with Respect to Mixture Means

For the component means $\{\mu_m\}_{m=1}^{K-1}$, the application-stage per-draw gradient for type k with respect to μ_m follows the general form in Equation (7). For draw r , we can write:

$$g_{ir,k}^A(\mu_m) = \frac{1}{\lambda} \left[\frac{\partial e_{ir,k}(A_i)}{\partial \mu_m} - \sum_a P_{A,irk}(a) \frac{\partial e_{ir,k}(a)}{\partial \mu_m} \right],$$

where $e_{ir,k}(a)$ denotes the application-stage expected utility for type k and application vector a , and $P_{A,irk}(a)$ is the corresponding application probability for type k .

For types $k < K$, the expected utility $e_{ir,k}(a)$ contains a term that is linear in μ_k with coefficient $\pi_{ia} P_{E,irk}$, so the derivative with respect to μ_m is:

$$\frac{\partial e_{ir,k}(a)}{\partial \mu_m} = \begin{cases} \pi_{ia} P_{E,irk}, & \text{if } m = k, \\ 0, & \text{if } m \neq k. \end{cases}$$

For type K , the mean μ_K is not a free parameter but is pinned down by the mean-zero restriction $E[\theta] = 0$. As noted above, this implies:

$$\mu_K = - \frac{\sum_{k'=1}^{K-1} p_{k'} \mu_{k'}}{p_K},$$

so that for $m < K$,

$$\frac{\partial \mu_K}{\partial \mu_m} = - \frac{p_m}{p_K}.$$

In turn, the μ_K -dependent part of the expected utility for type K is proportional to $\pi_{ia} P_{E,i r K} \mu_K$, so by the chain rule we obtain:

$$\begin{aligned} \frac{\partial e_{ir,K}(a)}{\partial \mu_m} &= \pi_{ia} P_{E,i r K} \frac{\partial \mu_K}{\partial \mu_m} \\ &= -\pi_{ia} P_{E,i r K} \frac{p_m}{p_K}, \quad m = 1, \dots, K-1. \end{aligned}$$

There is no free parameter μ_K , so we do not compute a gradient with respect to μ_K itself.

Substituting these derivatives into the general application-stage gradient expression above yields $g_{ir,k}^A(\mu_m)$ for each type k and mean parameter μ_m . The corresponding enrollment-stage gradients with respect to μ_m are analogous, obtained by replacing $e_{ir,k}(a)$ with the enrollment utility index v_{ijr} and applying the same derivative rules. For brevity, we do not display them here explicitly.

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